

Anexo 02 — Modelo fisico base (MJCF)

Archivo: Cuerpo/DUM4.xml

Para que sirvio:

Modelo MJCF base exportado de Fusion360 (plugin ACDC4Robot) y corregido a mano: unidades a radianes, limites de articulaciones, propiedades de actuadores y la restriccion leva->dedos. Es la version sin los elementos de la tarea de agarre, usada para el entrenamiento del seguimiento con la cabeza.

```
1 | <mujoco model="DUM4">
2 |   <compiler angle="radian" />
3 |   <option integrator="implicit" timestep="0.005" />
4 |   <visual>
5 |     <global offwidth="1280" offheight="720" />
6 |   </visual>
7 |
8 |   <asset>
9 |     <mesh name="Base_link" file="meshes/Base_link.stl" scale="0.001 0.001 0.001" />
10 |    <mesh name="BaseHip_link" file="meshes/BaseHip_link.stl" scale="0.001 0.001 0.001" />
11 |    <mesh name="FullBody_link" file="meshes/FullBody_link.stl" scale="0.001 0.001 0.001" />
12 |    <mesh name="BodyLeftShoulder_Link" file="meshes/BodyLeftShoulder_Link.stl" scale="0.001 0.001 0.001" />
13 |    <mesh name="BodyShoulderRight_Link" file="meshes/BodyShoulderRight_Link.stl" scale="0.001 0.001 0.001" />
14 |    <mesh name="RightShoulderArm_Link" file="meshes/RightShoulderArm_Link.stl" scale="0.001 0.001 0.001" />
15 |    <mesh name="LeftShoulderArm_Link" file="meshes/LeftShoulderArm_Link.stl" scale="0.001 0.001 0.001" />
16 |    <mesh name="ForearmLeft_link" file="meshes/ForearmLeft_link.stl" scale="0.001 0.001 0.001" />
17 |    <mesh name="ForearmRight_link" file="meshes/ForearmRight_link.stl" scale="0.001 0.001 0.001" />
18 |    <mesh name="WristRight_link" file="meshes/WristRight_link.stl" scale="0.001 0.001 0.001" />
19 |    <mesh name="FingersLeverRight_link" file="meshes/FingersLeverRight_link.stl" scale="0.001 0.001 0.001" />
20 |    <mesh name="RigthTopFinger_link" file="meshes/RigthTopFinger_link.stl" scale="0.001 0.001 0.001" />
21 |    <mesh name="RightBotFinger_link" file="meshes/RightBotFinger_link.stl" scale="0.001 0.001 0.001" />
22 |    <mesh name="WristLeft_link" file="meshes/WristLeft_link.stl" scale="0.001 0.001 0.001" />
23 |    <mesh name="FingersLeverLeft_link" file="meshes/FingersLeverLeft_link.stl" scale="0.001 0.001 0.001" />
24 |    <mesh name="LeftTopFinger_link" file="meshes/LeftTopFinger_link.stl" scale="0.001 0.001 0.001" />
25 |    <mesh name="LeftBotFinger_link" file="meshes/LeftBotFinger_link.stl" scale="0.001 0.001 0.001" />
26 |    <mesh name="Neck_link" file="meshes/Neck_link.stl" scale="0.001 0.001 0.001" />
27 |    <mesh name="HeadBase_link" file="meshes/HeadBase_link.stl" scale="0.001 0.001 0.001" />
28 |    <mesh name="LenteExt_link" file="meshes/LenteExt_link.stl" scale="0.001 0.001 0.001" />
29 |    <mesh name="LenteInt_link" file="meshes/LenteInt_link.stl" scale="0.001 0.001 0.001" />
30 |  </asset>
31 |  <worldbody>
32 |    <light directional="true" pos="-0.5 0.5 3" dir="0 0 -1" />
33 |    <geom pos="0 0 0" size="1 1 1" type="plane" rgba="1 0.83 0.61 0.5" />
34 |    <!-- Target visual para RL head-tracking (mocap = pose seteada desde codigo) -->
35 |    <body name="target" mocap="true" pos="0 0 1.0">
36 |      <geom name="target_geom" type="sphere" size="0.025" rgba="1 0.1 0.1 1" contype="0" conaffinity="0" />
37 |    </body>
38 |    <body name="Base_link" pos="-0.49041465089346836 0.6163554147407629 0.09256743876909335"
euler="2.192008610302898e-17 2.1137487076796193e-17 -1.5707963267949199">
39 |      <geom name="Base_link_geom" type="mesh" mesh="Base_link" pos="0 0 0" euler="0 0 0" />
40 |      <inertial mass="2.5686560790498238" pos="0.4059758533032169 0.6030766279468239 -0.05259420379589351"
fullinertia="0.1162260390457559 0.4484597933046195 0.5468615218419377 0.00021367853479986663 0.013204837742259792
-0.0001988317425628886" />
41 |      <body name="BaseHip_link" pos="0.6833656443798684 0.38691435800750773 0.02959497105331984"
euler="6.123233995761558e-17 -1.3506756093454076e-14 1.5707963267949199">
42 |        <!-- <joint name="BaseHip_joint" type="hinge" axis="-3.702493273007566e-58 1.0
1.3484836007351056e-14" pos="0.21545729288597631 0.3064582296391153 -0.0396004098224091" limited="true" range="-0.0000001
0.0000001" frictionloss="0.1"/> -->
43 |        <geom name="BaseHip_link_geom" type="mesh" mesh="BaseHip_link" pos="0 0 0" euler="0 0 0" />
44 |        <inertial mass="0.8960915657403068" pos="0.21384545840294725 0.28547506459516403
-0.028631766992634693" fullinertia="0.0004072702169369141 0.000707340860722314 0.000496468604737454 6.3409287313042134e-06
-2.5719708909663157e-06 -1.867087549890158e-05" />
45 |        <body name="FullBody_link" pos="0.492500000000000066 0.19200000000000003 0.16054771464531606"
euler="-0.0 0.0 4.2193972572360445e-42">
46 |          <joint name="HipBody_joint" type="hinge" axis="1.0 5.406725158324904e-31
-4.0094852880505995e-17" pos="-0.3067067071140243 0.09941022963911494 -0.19996812446772536" limited="true"
range="-0.523598776 0.523598776" damping="40.0" frictionloss="0.1"/>
47 |          <geom name="FullBody_link_geom" type="mesh" mesh="FullBody_link" pos="0 0 0" euler="0 0 0" />
48 |          <inertial mass="0.03077775275728115" pos="-0.2748562047070093 0.07604771905982315
-0.042809377340932674" fullinertia="0.05913846104456666 0.06751063222309828 0.0314000156710864 3.531098098097749e-05
-0.00027353184418924046 0.0017686717833686899" />
49 |          <body name="BodyShoulderLeft_Link" pos="0.0 -0.00049999999999999894 3.552713678800501e-17"
euler="-0.0 0.0 7.888609052216694e-31">
50 |            <joint name="BodyShoulderLeft_joint" type="hinge" axis="-7.888609052210118e-31 -1.0
-1.3484836007351056e-14" pos="-0.23487070711402433 0.08669122963911181 0.0344728753227442" limited="true" range="-0.1309
0.1309" damping="5.0" frictionloss="0.05"/>
51 |            <geom name="BodyShoulderLeft_Link_geom" type="mesh" mesh="BodyLeftShoulder_Link" pos="0 0 0"
euler="0 0 0" />
52 |            <inertial mass="0.001580164403355698" pos="-0.23097897492018266 0.07102238334903822
0.037812155469283126" fullinertia="0.00010342997815528804 0.00012247959556750676 5.5657869369753866e-05
3.1812903011561105e-06 -4.646173970512077e-06 -7.820179979611136e-07" />
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53 | <body name="LeftShoulderArm_Link" pos="0.00500000000000012795 -0.001318155121034277
-0.0011714595677372942" euler="-1.8251529652410896e-17 4.019437521819588e-15 8.630902712646249e-18">
54 | <joint name="LeftShoulderArm_joint" type="hinge" axis="1.0 -8.630902712646497e-18
3.979342668939082e-15" pos="-0.21423370711402578 0.07000038476014589 0.0500003351000106" limited="true" range="-0.087266463
0.087266463" damping="5.0" frictionloss="0.05"/>
55 | <geom name="LeftShoulderArm_Link_geom" type="mesh" mesh="LeftShoulderArm_Link" pos="0 0
0" euler="0 0 0" />
56 | <inertial mass="0.006044348563915437" pos="-0.14775520501081005 0.07000069785250757
-0.03493149873121201" fullinertia="0.006742433433134626 0.006858876869892644 0.0005583571131118684 -3.4859104103944944e-09
0.0007244169671068692 2.7789685096562462e-08" />
57 | <body name="LeftForearm_link" pos="0.0012660897788712333 8.260059303211165e-15
-2.948752353404416e-15" euler="-0.0 -2.8656915481489102e-15 -8.630902712509018e-18">
58 | <joint name="LeftForearm_joint" type="hinge" axis="-1.0 1.3747904240618293e-28
-1.113651120790172e-15" pos="-0.13399979689289668 0.07000038476013982 -0.11321466489998573" damping="0.75"
frictionloss="0.01"/>
59 | <geom name="LeftForearm_link_geom" type="mesh" mesh="ForearmLeft_link" pos="0 0 0"
euler="0 0 0" />
60 | <inertial mass="0.00760399800858365" pos="-0.15628264744397327 -0.025728185625634687
-0.11508012737492908" fullinertia="0.0036280202441539636 0.0010097063275369732 0.003658026145208641 7.969107417180748e-05
5.0947784061571735e-06 -0.00015416807508479666" />
61 | <body name="WristLeft_link" pos="-0.01999999999999993 1.2434497875801754e-16 0.0"
euler="-1.9428902930935822e-16 0.0 -1.3821853194088672e-28">
62 | <joint name="LeftWrist_joint" type="hinge" axis="1.5777218104420236e-29 -1.0
-1.3697376566312825e-14" pos="-0.13443279689289678 -0.10388761523986029 -0.1132796648999881" damping="0.75"
frictionloss="0.01"/>
63 | <geom name="WristLeft_link_geom" type="mesh" mesh="WristLeft_link" pos="0 0 0"
euler="0 0 0" />
64 | <inertial mass="0.0010523058119647006" pos="-0.13094086013592526
-0.11928841654945467 -0.11223969951542469" fullinertia="0.00013484065595188027 0.00017430131996684645
0.00015570833627770447 -7.063156137799158e-06 -4.330604677182016e-06 -2.2987911685198255e-06" />
65 | <body name="LeftFingersLever_link" pos="-0.04648622952275133 0.02574992388730566
-0.0019991037873249694" euler="1.4966003627485e-15 0.0 -7.888609052263423e-31">
66 | <joint name="LeftLever_Slider" type="slide" axis="1.6566079009641248e-29
-1.0 -1.2200776203564325e-14" pos="-0.08793256737014545 -0.1418875391271661 -0.1112805611126631" limited="true" range="0.0
0.02" damping="0.75" frictionloss="0.01"/>
67 | <geom name="LeftFingersLever_link_geom" type="mesh"
mesh="FingersLeverLeft_link" pos="0 0 0" euler="0 0 0" />
68 | <inertial mass="0.0002502162568032953" pos="-0.08793286311977296
-0.1580316237704062 -0.11128018896759909" fullinertia="1.4346397986760343e-07 8.059226657580222e-08 2.2115726436549438e-07
-2.6977496231812774e-14 -5.458814349447949e-11 -6.865250339310314e-15" />
69 | </body>
70 | <body name="LeftTopFinger_link" pos="-0.09150000058017238 0.0059999999999994508
-0.0070000005854844353" euler="-0.0 -9.8362225043893e-17 1.57772181044155e-29">
71 | <joint name="LeftFingerTop_joint" type="hinge" axis="-1.0
1.5254098764959316e-29 -1.0152888982857827e-15" pos="-0.0434327963127245 -0.15081761523985507 -0.0862976590451443"
limited="true" range="0.0 0.34906585" damping="0.5" frictionloss="0.01"/>
72 | <geom name="LeftTopFinger_link_geom" type="mesh" mesh="LeftTopFinger_link"
pos="0 0 0" euler="0 0 0" />
73 | <inertial mass="0.00017504820565798226" pos="-0.043439917104673695
-0.19042347475958155 -0.0868188581695989" fullinertia="0.00011707275115569078 1.0569157708168285e-05 0.00011288264167109947
-5.87246265930031415e-08 2.8321002246495928e-08 -1.3696662666319162e-05" />
74 | </body>
75 | <body name="LeftBotFinger_link" pos="-0.091499999999999932 0.0059999999999994703
-0.006992418275234122" euler="-0.0 0.0 1.57772181044155e-29">
76 | <joint name="LeftFingerBot_joint" type="hinge" axis="-1.0
1.5254098764959316e-29 -1.113651120790172e-15" pos="-0.04293279689289753 -0.15081761523985474 -0.12626224662475452"
limited="true" range="-0.34906585 0.0" damping="0.5" frictionloss="0.01"/>
77 | <geom name="LeftBotFinger_link_geom" type="mesh" mesh="LeftBotFinger_link"
pos="0 0 0" euler="0 0 0" />
78 | <inertial mass="0.00032877369249164234" pos="-0.042938369298710595
-0.19025542357391964 -0.12573014078361325" fullinertia="0.00022143789541416437 5.882586795043846e-05 0.0002523423378452169
7.83221744205041e-08 1.9216897559315403e-08 2.508596718914621e-05" />
79 | </body>
80 | </body>
81 | </body>
82 | </body>
83 | </body>
84 | <body name="RightBodyShoulder_Link" pos="0.0 -0.0004999999999999905 -4.263256414560601e-16"
euler="4.728783881889385e-15 0.0 1.019463261665369e-31">
85 | <joint name="RightBodyShoulder_joint" type="hinge" axis="-5.406725158324904e-31 -1.0
-8.75605212546167e-15" pos="-0.3151297071140243 0.08669122963911187 0.03447287553227451" limited="true" range="-0.087266463
0.087266463" damping="5.0" frictionloss="0.05"/>
86 | <geom name="RightBodyShoulder_Link_geom" type="mesh" mesh="BodyShoulderRight_Link" pos="0 0
0" euler="0 0 0" />
87 | <inertial mass="0.0015801644033557983" pos="-0.31902102507983776 0.0710223833490348
0.03781215546928169" fullinertia="0.00010342997815538518 0.00012247959556840188 5.565786937072012e-05
-3.181290301413695e-06 4.64617397038025e-06 -7.820179980065322e-07" />
88 | <body name="RightShoulderArm_Link" pos="-0.0049999999999999787 8.881784197001253e-17
1.3500311979441904e-15" euler="-6.626431603856499e-29 -3.986888380656777e-15 -3.4460753534106624e-29">
89 | <joint name="RightShoulderArm_joint" type="hinge" axis="1.0 3.470987982972452e-29
-4.026983233537283e-15" pos="-0.3397657071140243 0.06868222963911166 0.0488288755322743" limited="true"
range="-0.261799388 0.261799388" damping="5.0" frictionloss="0.05"/>
90 | <geom name="RightShoulderArm_Link_geom" type="mesh" mesh="RightShoulderArm_Link" pos="0
0 0" euler="0 0 0" />
91 | <inertial mass="0.006044348406701217" pos="-0.4012797494257568 0.06868254273148353
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-0.036102959407607306" fullinertia="0.006742433319325109 0.006855275668464347 0.0005547560251000772 3.4565310156444675e-09
-0.0007208413951458895 2.7789684143787037e-08" />
  92 | <body name="RightForearm_link" pos="0.00013390950585971952 -0.0013181551210373144
-0.0011714595677335992" euler="-0.0 0.0 -5.5059527148838425e-31">
  93 | <joint name="RightForearm_joint" type="hinge" axis="-1.0 -3.5260475101212904e-29
4.026983233537283e-15" pos="-0.4163996166198847 0.07000038476015039 -0.11321466489999173" limited="true" range="-0.34906585
0.698131701" damping="0.75" frictionloss="0.01"/>
  94 | <geom name="RightForearm_link_geom" type="mesh" mesh="ForearmRight_link" pos="0 0 0"
euler="0 0 0" />
  95 | <inertial mass="0.007580802843149587" pos="-0.3940021843225092 -0.025789916933327592
-0.11508573340750891" fullinertia="0.003625282012670414 0.001004896002765601 0.00365039126096615 -8.572215713482709e-05
-5.4252145051112465e-06 -0.00015409686018611125" />
  96 | <body name="RightWrist_link" pos="0.031099999999999354 0.00100000000000029985
-2.913225216616411e-15" euler="2.3314683517127853e-15 0.0 3.4709879829723886e-29">
  97 | <joint name="RightWrist_joint" type="hinge" axis="3.470987982972452e-29 1.0
6.424583773748951e-15" pos="-0.42706761661988407 -0.10488761523985289 -0.11327966489999021" damping="0.75"
frictionloss="0.01"/>
  98 | <geom name="RightWrist_link_geom" type="mesh" mesh="WristRight_link" pos="0 0 0"
euler="0 0 0" />
  99 | <inertial mass="0.001552756438192612" pos="-0.4305589870484671
-0.12018641955759513 -0.11224742302493411" fullinertia="0.00013512664414972597 0.0001743909362236108 0.00015590432307399096
7.394754135254912e-06 4.342023717063553e-06 -2.383774291398711e-06" />
  100 | <body name="RightFingersLever_link" pos="0.039886229522752434
0.0014783163197503947 -0.0012398490413806586" euler="-0.010902936554985417 1.2246467991471227e-16 4.026354632581604e-15">
  101 | <joint name="RightLever_Slider" type="slide" axis="-3.982450237386041e-15
-1.0000004394448656 2.746815654895707e-07" pos="-0.467066846142637 -0.11688781938049393 -0.11327994404304859"
limited="true" range="0.0 0.02" damping="0.75" frictionloss="0.01"/>
  102 | <geom name="RightFingersLever_link_geom" type="mesh"
mesh="FingersLeverRight_link" pos="0 0 0" euler="0 0 0" />
  103 | <inertial mass="0.0002502162568032968" pos="-0.46706713502130953
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-2.6977652905654876e-14 5.4588121651559864e-11 -6.8654913564264514e-15" />
  104 | </body>
  105 | <body name="RigthTopFinger_link" pos="0.05500000002874209 0.0
-3.552713678800501e-17" euler="-0.0 1.5387337373564732e-16 -3.3721307449609837e-29">
  106 | <joint name="RightFingerTop_joint" type="hinge" axis="-1.0
-2.4883118759228663e-29 3.873109859801636e-15" pos="-0.48156761664862613 -0.14581761523985298 -0.09329766489999049"
limited="true" range="0.0 0.34906585" damping="0.5" frictionloss="0.01"/>
  107 | <geom name="RigthTopFinger_link_geom" type="mesh" mesh="RigthTopFinger_link"
pos="0 0 0" euler="0 0 0" />
  108 | <inertial mass="0.0017504820557522993" pos="-0.4815600828953232
-0.1854234747802852 -0.09381886402104374" fullinertia="0.00011707275098959795 1.0569157699652328e-05 0.00011288264151040553
5.872426590600409e-08 -2.83210022383226e-08 -1.3696662693378802e-05" />
  109 | </body>
  110 | <body name="RightBotFinger_link" pos="0.055000000000000146 0.0016019936645571776
-0.0008101076126539652" euler="-0.0 1.5387337373564811e-16 -3.470987982972388e-29">
  111 | <joint name="RightFingerBot_joint" type="hinge" axis="-1.0
-2.587169113934271e-29 3.873109859801635e-15" pos="-0.4820676166198843 -0.14581760890440992 -0.13326255728733652"
limited="true" range="-0.34906585 0.0" damping="0.5" frictionloss="0.01"/>
  112 | <geom name="RightBotFinger_link_geom" type="mesh" mesh="RightBotFinger_link"
pos="0 0 0" euler="0 0 0" />
  113 | <inertial mass="0.003287736926194663" pos="-0.4820616307012917
-0.18525542355744726 -0.13273014663702865" fullinertia="0.00022143789564719324 5.882586796765595e-05 0.00025234233808858243
-7.832217439595916e-08 -1.9216897555976945e-08 2.5085967168581173e-05" />
  114 | </body>
  115 | </body>
  116 | </body>
  117 | </body>
  118 | </body>
  119 | <body name="Neck_link" pos="1.4210854715202004e-16 0.00036714200512303567 -0.000262232066032162"
euler="-5.988489307583366e-17 1.1542917984037797e-15 -1.4409821234723413e-31">
  120 | <joint name="Neck_joint" type="hinge" axis="-1.0 -6.8717180415736575e-31
-1.1141969455232755e-15" pos="-0.2994997071140246 0.07644508763398802 0.10105010759830632" limited="true"
range="-0.872664626 1.221730476" damping="5.0" frictionloss="0.1"/>
  121 | <geom name="Neck_link_geom" type="mesh" mesh="Neck_link" pos="0 0 0" euler="0 0 0" />
  122 | <inertial mass="0.003004544673814925" pos="-0.27396409671398037 0.07644546299124172
0.14604028758513957" fullinertia="0.0004813487439827985 0.0005288328590369821 0.00012405625886341458
-2.1776938587108386e-10 1.0117082966391687e-06 -1.4063382201748417e-10" />
  123 | <body name="HeadBase_link" pos="-2.7749467641058345e-07 -0.00036714200512347975
0.0002617680788097232" euler="-2.2204469179127474e-16 1.5153644512849699e-06 1.200032827176549e-22">
  124 | <joint name="HeadBase_joint" type="hinge" axis="1.0000000000018825
2.8079661542830124e-20 -4.846355475991192e-07" pos="-0.29799971873287356 0.0768122296391002 0.19078788794153526"
limited="true" range="-0.698131701 0.698131701" damping="5.0" frictionloss="0.1"/>
  125 | <geom name="HeadBase_link_geom" type="mesh" mesh="HeadBase_link" pos="0 0 0" euler="0 0
0" />
  126 | <inertial mass="0.008292197938464645" pos="-0.27341672481886703 0.08325319774714132
0.21831365816936063" fullinertia="0.0006128769725709957 0.0008578490533403282 0.0010644576576086845 -1.2084540966775883e-05
1.085184479466515e-05 -7.362870407013952e-05" />
  127 | <body name="LenteExt_link" pos="0.0004999999999997585 0.0009999999999998899
-1.4210854715202004e-16" euler="5.551115123126413e-17 8.264055774062899e-17 1.0648908069593082e-21">
  128 | <joint name="HeadRotation_joint" type="hinge" axis="4.84635547516478e-07
-1.365574320289045e-14 1.0000000000018827" pos="-0.2744997566170123 0.07581222963910991 0.21578792431025356" limited="true"
range="-2.79253 2.79253" damping="5.0" frictionloss="0.1"/>
  129 | <geom name="LenteExt_link_geom" type="mesh" mesh="LenteExt_link" pos="0 0 0"
euler="0 0 0"/>
```

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130 |                                     <!-- Site para calcular theta: posicionado en el centro visual del lente (ipos del
CoM).
131 |                                     Resuelve el "desfase constante" porque el origen real del rayo de mirada
132 |                                     coincide con el lente, no con body.xpos que esta a 0.37m de distancia.
133 |                                     Invisible al render (alpha=0) - ya validado el axis, no hace falta verlo. -->
134 |                                     <site name="lens_center" pos="-0.274 0.013 0.220" size="0.005" rgba="1 1 0 0.0"
type="sphere"/>
135 |                                     <inertial mass="0.04614137041683628" pos="-0.2739713756270979 0.01302389740638243
0.22079805785885293" fullinertia="0.08560328250932785 0.045553726512780016 0.10676785371969727 0.0003031396090717247
9.679865428052625e-05 -0.011028121611163578" />
136 |                                     <body name="LenteInt_link" pos="1.4210854715202004e-16 -0.014999999999999995
-6.394884621840901e-16" euler="4.9789962505144925e-17 6.123201175701337e-17 6.123233995736765e-17">
137 |                                     <joint name="LenteExt_joint" type="slide" axis="6.121215846663226e-17 1.0
1.3661464478465951e-14" pos="-0.2734467480021667 -0.08818777036089008 0.21010292590593704" limited="true" range="0.0 0.021"
damping="5.0" frictionloss="0.01"/>
138 |                                     <geom name="LenteInt_link_geom" type="mesh" mesh="LenteInt_link" pos="0 0 0"
euler="0 0 0" />
139 |                                     <inertial mass="0.003021264973037482" pos="-0.27344687236060033
-0.08907562592418654 0.20923732950881085" fullinertia="8.50114671194762e-05 0.00014657747592818077 9.180919722068854e-05
-1.8089708046838876e-19 6.186000055722115e-18 -1.3309448564103802e-06" />
140 |                                     </body>
141 |                                     </body>
142 |                                     </body>
143 |                                     </body>
144 |                                     </body>
145 |                                     </body>
146 |                                     </body>
147 |                               </worldbody>
148 |                               <contact>
149 |                                   <!-- Excluye colisiones internas del craneo que bloquean al HeadRot.
150 |                                   Las mallas de LenteExt rozan con HeadBase/Neck cuando rotan; lo cual frena el yaw.
151 |                                   Se mantienen las colisiones del lente con el resto del mundo. -->
152 |                                   <exclude body1="LenteExt_link" body2="HeadBase_link"/>
153 |                                   <exclude body1="LenteExt_link" body2="Neck_link"/>
154 |                                   <exclude body1="LenteInt_link" body2="HeadBase_link"/>
155 |                               </contact>
156 |                               <equality>
157 |                                   <joint name="ELSliderTop_joint" joint1="LeftFingerTop_joint" joint2="LeftLever_Slider" polycoef="0 34.906585
0 0 0"/>
158 |                                   <joint name="ELSliderBot_joint" joint1="LeftFingerBot_joint" joint2="LeftLever_Slider" polycoef="0
-34.906585 0 0 0"/>
159 |                                   <joint name="ERSliderTop_joint" joint1="RightFingerTop_joint" joint2="RightLever_Slider" polycoef="0
34.906585 0 0 0"/>
160 |                                   <joint name="ERSliderBot_joint" joint1="RightFingerBot_joint" joint2="RightLever_Slider" polycoef="0
-34.906585 0 0 0"/>
161 |                               </equality>
162 |                               <actuator>
163 |                                   <!-- TORSO Y CUELLO (Basados en DS3218 - High Torque) -->
164 |                                   <!-- Torque: 21.5 kg-cm = 2.11 N-m. Velocidad: 0.14s/60deg -->
165 |                                   <!-- <position name="act_BaseHip" joint="BaseHip_joint" ctrlrange="-0.0001 0.0001" kv="0" kp="0"
forcerange="-2.11 2.11" gear="1" /> -->
166 |                                   <position name="act_HipBody" joint="HipBody_joint" ctrlrange="-0.523 0.523" kv="70" kp="800"
forcerange="-12.11 12.11" gear="3" />
167 |                                   <position name="act_Neck" joint="Neck_joint" ctrlrange="-0.872 1.221" kv="24" kp="3000" forcerange="-2.11
2.11"/>
168 |                                   <position name="act_HeadBase" joint="HeadBase_joint" ctrlrange="-0.698 0.698" kv="18" kp="600"
forcerange="-2.11 2.11" />
169 |                                   <position name="act_HeadRot" joint="HeadRotation_joint" ctrlrange="-2.79253 2.79253" kv="38" kp="3000"
forcerange="-2.11 2.11"/>
170 |                                   <!-- BRAZOS Y HOMBROS (Basados en MG996R - Metal Gear) -->
171 |                                   <!-- Torque: 11 kg-cm = 1.08 N-m. Velocidad: 0.15s/60deg -->
172 |                                   <!-- Actuadores nuevos para los body shoulders (movimiento ligero hombros pegado al torso) -->
173 |                                   <position name="act_BodyShoulderLeft" joint="BodyShoulderLeft_joint" ctrlrange="-0.1309 0.1309" kv="30"
kp="800" forcerange="-1.08 1.08"/>
174 |                                   <position name="act_BodyShoulderRight" joint="RightBodyShoulder_joint" ctrlrange="-0.1309 0.1309" kv="30"
kp="800" forcerange="-1.08 1.08"/>
175 |                                   <position name="act_LeftShoulderArm" joint="LeftShoulderArm_joint" ctrlrange="-0.087 0.087" kv="16"
kp="500" forcerange="-2.11 2.11"/>
176 |                                   <position name="act_RightShoulderArm" joint="RightShoulderArm_joint" ctrlrange="-0.261 0.261" kv="16"
kp="500" forcerange="-2.11 2.11"/>
177 |                                   <position name="act_LeftForearm" joint="LeftForearm_joint" ctrlrange="-1.57 1.57" kv="20" kp="1000"
forcerange="-2.11 2.11"/>
178 |                                   <position name="act_RightForearm" joint="RightForearm_joint" ctrlrange="-0.349 0.698" kv="20" kp="1400"
forcerange="-2.11 2.11"/>
179 |                                   <!-- MUÑECAS Y MECANISMOS (Basados en MG90S - Micro Metal Gear) -->
180 |                                   <!-- Torque: 2.2 kg-cm = 0.21 N-m. Velocidad: 0.08s/60deg -->
181 |                                   <position name="act_LeftWrist" joint="LeftWrist_joint" ctrlrange="-1.57 1.57" kv="20" kp="900"
forcerange="-0.21 0.21"/>
182 |                                   <position name="act_RightWrist" joint="RightWrist_joint" ctrlrange="-1.57 1.57" kv="20" kp="900"
forcerange="-0.21 0.21"/>
183 |                                   <!-- MECANISMOS DE PINZA (Accionados por MG90S via Leva) -->
184 |                                   <position name="act_LeftLever_Slider" joint="LeftLever_Slider" ctrlrange="0.0 0.02" kv="20" kp="1400"
forcerange="-1.08 1.08"/>
185 |                                   <position name="act_RightLever_Slider" joint="RightLever_Slider" ctrlrange="0.0 0.02" kv="20" kp="1400"
forcerange="-1.08 1.08"/>

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186 |         <!-- LENTE INTERNO (Enfoque animatronico, slide del lente interior) -->
187 |         <position name="act_LenteExt" joint="LenteExt_joint" ctrlrange="0.0 0.021" kv="5" kp="800"
forcerange="-0.21 0.21"/>
188 |         </actuator>
189 | </mujoco>
```